
SEMANTIC FORCE CLOSURE: TASK-LEGAL CONTACT CERTIFICATES FOR ROBOTIC GRASPING

Anonymous authors

Paper under double-blind review

ABSTRACT

Robotic grasping often treats semantics as a pre-filter or post-filter around a geometric force-closure test. That separation is unsafe for task-oriented manipulation: a grasp can be geometrically force closed because it touches a blade, rim, screen, nozzle, stem, or tip that the task explicitly forbids. We introduce Semantic Force Closure (SFC), a certificate that deletes task-forbidden contact-role wrench generators before checking whether the origin remains in the convex hull of the surviving generators. In a deterministic planar contact study over six manipulation tasks and 720 trials, geometric force closure succeeds in all trials, while SFC succeeds in only 65.4%. The resulting 34.6% geometric optimism gap exposes failures that semantic-only contact acceptance and post-hoc filtering both miss. SFC is intentionally a small mechanism: it changes the certificate witness, not the perception backbone.

1 INTRODUCTION

Task-oriented grasping has two obligations that are usually separated. It must be mechanically stable, and it must touch the object in task-legal places. Classical force closure addresses the first obligation with wrench geometry; semantic grasping and affordance grounding address the second with part labels, language-conditioned rankings, or masks. The problem is that the two modules do not commute. A certificate found using all contacts can rely on a forbidden part, and removing that part after the fact may destroy the wrench balance.

We call the desired certificate Semantic Force Closure. Given contact roles and a task legality set, SFC removes every wrench generator attached to a forbidden role before running the force-closure feasibility check. This turns semantics from a score into a constraint on the certificate witness.

2 RELATED WORK

The closest mechanical literature studies force closure, grasp quality, and convex or LMI-style grasp analysis. The closest semantic literature studies functional grasping, affordance grounding, contact semantic maps, and task-oriented dexterous grasp datasets (Authors, 2023; 2025a;b; 2004; 2024; 1992; 2012; 2011). These works make semantic contact selection increasingly strong, but the common pattern is still to rank candidates before a geometric check or filter them after a geometric witness has already been chosen. SFC uses the same underlying mechanics but changes the order of operations.

3 SEMANTIC FORCE CLOSURE

Let each contact role be r_i and each contact have a linearized friction cone with wrench generators $w_{ij} \in \mathbb{R}^3$. Let $A(\tau)$ be the set of roles allowed by task τ . The SFC certificate is feasible when there are nonnegative coefficients λ_{ij} such that

$$\sum_{i:r_i \in A(\tau)} \sum_j \lambda_{ij} w_{ij} = 0, \quad \sum_{i:r_i \in A(\tau)} \sum_j \lambda_{ij} = 1.$$

Table 1: Rates in percent. FC any is ordinary geometric force closure. SFC is force closure after task-forbidden contact roles are removed.

Scenario	FC any	SFC	FC-not-SFC	Posthoc miss
knife_pass	100.0	0.0	100.0	0.0
mug_serve	100.0	100.0	0.0	100.0
phone_pick	100.0	100.0	0.0	100.0
spray_bottle_use	100.0	100.0	0.0	100.0
apple_pick	100.0	89.2	10.8	89.2
scissors_pass	100.0	3.3	96.7	3.3
All	100.0	65.4	34.6	65.4

This is the usual convex-hull force-closure test after semantic deletion. Because the allowed generator set is a subset of the geometric generator set, SFC implies geometric force closure in the same model. The reverse implication does not hold.

4 EXPERIMENT

We simulate planar rigid-body contacts for six task families: knife handoff, mug serving, phone pickup, spray bottle use, apple picking, and scissors passing. Each object has allowed roles and forbidden roles. Some trials bias allowed contacts into mechanically poor arcs while forbidden contacts remain mechanically attractive. For every trial we compare geometric force closure using all contacts, SFC using only allowed contacts, semantic-only acceptance, and post-hoc acceptance of the geometric witness.

5 RESULTS

Table 1 reports rates over 720 deterministic trials. Geometric force closure succeeds in every trial, but SFC succeeds in 65.4%. The geometric optimism gap is 34.6%, and post-hoc filtering misses alternative allowed-only certificates in 65.4% of all trials. The adversarial monotonicity check found zero cases where SFC succeeded while geometric force closure failed.

6 DISCUSSION AND LIMITATIONS

SFC is not a replacement for perception or trajectory optimization. It assumes contact-role labels and uses a planar linearized friction model. Its value is diagnostic: it identifies when a grasp proposal is mechanically stable only because it uses a task-illegal contact. A full system should combine SFC with uncertainty over roles, tactile verification, and multi-step planning.

7 CONCLUSION

Semantic grasping should change the contact certificate, not only the ranking around it. Semantic Force Closure is a small certificate-level mechanism that exposes a concrete optimism gap between geometric stability and task-legal stability.

REFERENCES

- OpenAlex Indexed Authors. Grasp synthesis of polygonal objects using a three-fingered robot hand. *The International Journal of Robotics Research*, 1992.
- OpenAlex Indexed Authors. On quality functions for grasp synthesis, fixture planning, and coordinated manipulation. *IEEE Transactions on Automation Science and Engineering*, 2004.
- OpenAlex Indexed Authors. Affordance based part recognition for grasping and manipulation. *venue unavailable*, 2011.

108 OpenAlex Indexed Authors. Semantic grasping: Planning robotic grasps functionally suitable for
109 an object manipulation task. *venue unavailable*, 2012.

110

111 OpenAlex Indexed Authors. Dexgraspnet: A large-scale robotic dexterous grasp dataset for general
112 objects based on simulation. *arXiv (Cornell University)*, 2023.

113

114 OpenAlex Indexed Authors. Robo-abc: Affordance generalization beyond categories via semantic
115 correspondence for robot manipulation. *Lecture notes in computer science*, 2024.

116

117 OpenAlex Indexed Authors. Learning 6-dof fine-grained grasp detection based on part affordance
grounding. *IEEE Transactions on Automation Science and Engineering*, 2025a.

118

119 OpenAlex Indexed Authors. Zerodexgrasp: Zero-shot task-oriented dexterous grasp synthesis with
120 prompt-based multi-stage semantic reasoning. *ArXiv.org*, 2025b.

121

122

123

124

125

126

127

128

129

130

131

132

133

134

135

136

137

138

139

140

141

142

143

144

145

146

147

148

149

150

151

152

153

154

155

156

157

158

159

160

161